Neil A. Ketteringham

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EDUCATION

University of California Irvine, School of Information & Computer Sciences *Graduation: Fall 2027*
Bachelor of Science in Computer Science GPA: 3.97

EXPERIENCE

Apache Texera (Incubating) *Jun 2026 - Present*
Undergraduate Researcher

- **Reduced database test suite runtime by ~28%** (5m 40s to 4m 05s) by refactoring Apache Texera's test harness into a JVM-wide singleton with **HikariCP** connection pooling and in-memory DDL caching, eliminating thread race conditions across 1,600+ assertions.
- **Engineered a multi-provider authentication system** in Scala/jOOQ and Angular by decoupling user identity into a unified auth_provider schema; built an idempotent OAuth 2.0/OIDC pipeline for ORCID integration with fallback email verification modals for restricted profiles.

Antikat Studios *Jan 2025 - Present*
Lead Programmer & Project Manager

- **Led a 12-person cross-functional team** to ship the game *"Japanese Salaryman"* (Unity/C#), driving product roadmaps and launching a Steam demo that achieved **2.7k wishlists** and **100% positive reviews**.
- **Engineered custom C# engine infrastructure**, including a polymorphic binary serializer with type-tagging to replace JSON parsing, and an interpreted cutscene DSL to automate narrative workflows.
- **Architected an event-driven scheduling pipeline** using SQLite for backend data storage paired with a custom Unity editor tool, enabling designers to dynamically balance complex game days.
- **Built a Go telemetry backend** with goroutine-based concurrent request handling and PostgreSQL persistence to capture real-time playtester data, deployed on a hardened Hetzner Linux server with systemd process management, Caddy-automated TLS, and schema migrations; replacing spreadsheet-only economy balancing with empirical playtest data.

Video Game Development Club (VGDC) @ UCI *Feb 2025 - Present*
Project Manager

- **Directed 10 concurrent cross-functional game development teams** (120+ total members) through complete 10-week development cycles from initial concept pitch to public showcase.
- **Designed and implemented an automated member-sorting algorithm** in Python, cutting team assignment processing time from 3 days to under an hour.
- **Organized and executed flagship quarterly showcases** and playtesting nights hosting 200+ attendees.

PROJECTS

Vulkan Renderer Engine | C++, Vulkan API, GLSL *May 2026*

- **Built a Vulkan graphics pipeline in C++,** manually configuring swapchain recreation, render passes, descriptor set layouts, and staging buffers to load and render textured 3D models.

N-Body Gravity Simulator | C++, OpenGL API, GLSL *Nov 2023*

- **Architected a real-time particle simulation i**n C++ and OpenGL with custom shader pipelines and VBO management to compute interactive gravitational physics.

SKILLS

Languages: C++, C#, Java, Scala, Python, TypeScript, JavaScript, SQL, Go, Lisp, Prolog
Systems & Graphics: Vulkan API, OpenGL, GLSL, Memory Management, Multi-threading / Concurrency
Frameworks & Web: Angular, Unity, REST APIs, WebSockets, OpenID, OAuth 2.0, JWT
Databases & Tools: PostgreSQL, jOOQ, HikariCP, Liquibase, sbt, Git, Docker, Linux